UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

FACULTAD DE CIENCIAS FÍSICAS

Teoría General de la Relatividad

2da. práctica
(Segundo semestre 2025)

Indicaciones Generales:
Materiales a utilizar: libros y apuntes.
Puntaje total: 20 puntos.
Las soluciones deben ser presentadas en el orden establecido de la práctica.
Fecha de entrega: **07/10/2025**.
Enviar la práctica resuelta al siguiente correo: **richard.avalos@unmsm.edu.pe**

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1. Utilizando la relación de cambio de base $\hat{e}'_a = \dfrac{\partial x^b}{\partial x'^a}\hat{e}_b$: (nueva = nueva, vieja = vieja) — covariantes

$$A'^a = \frac{\partial x'^a}{\partial x^b} A^b$$

a) Encuentre la matriz de transformación entre las bases de coordenadas cartesianas y las bases de coordenadas cilíndricas. **(1p)**

• Componentes covariantes:

Relación cartesianas - cilíndricas

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\operatorname{sen}\theta \\ z = z \end{cases}$$

cambio de base:

$$\hat{e}'_a = \frac{\partial x^b}{\partial x'^a}\hat{e}_b$$

$$\hat{e}_\rho = \frac{\partial x^b}{\partial \rho}\hat{e}_b \;,\quad \hat{e}_\theta = \frac{\partial x^b}{\partial \theta}\hat{e}_b \;,\quad \hat{e}_{z'} = \frac{\partial x^b}{\partial z'}\hat{e}_b$$

$dx = dx(\rho,\theta,z)$

$dy = \frac{\partial x}{\partial \rho}d\rho + \frac{\partial x}{\partial \theta}d\theta + \frac{\partial x}{\partial z}dz$

$dx = \cos\theta\, d\rho - \operatorname{sen}\theta\, d\theta$

$dy = \operatorname{sen}\theta\, d\rho + \cos\theta\, d\theta$

$dz = dz$

$\rho = \frac{x}{\cos\theta}$, pero ρ no debe depender de θ, son bases.

$ds^2 = dx^2 + dy^2 + dz^2 = d\rho^2 + \rho^2 d\varphi^2 + dz^2$

$ds^2 = g_{ij}\, dq^i dq^j$

$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $g_{ij} = e_i \cdot e_j$ $\begin{cases} \rho \\ \theta \\ z \end{cases}$

$$\hat{e}_\rho = \frac{\partial x}{\partial \rho}\hat{e}_x + \frac{\partial y}{\partial \rho}\hat{e}_y + \frac{\partial z}{\partial \rho}\hat{e}_z$$

$$\hat{e}_\rho = \cos\theta\,\hat{e}_x + \operatorname{sen}\theta\,\hat{e}_y$$

$$\hat{e}_\theta = \frac{\partial x}{\partial \theta}\hat{e}_x + \frac{\partial y}{\partial \theta}\hat{e}_y + \frac{\partial z}{\partial \theta}\hat{e}_z$$

$$\hat{e}_\theta = -r\operatorname{sen}\theta\,\hat{e}_x + r\cos\theta\,\hat{e}_y$$

$$\hat{e}_z = \frac{\partial x}{\partial z}\hat{e}_x + \frac{\partial y}{\partial z}\hat{e}_y + \frac{\partial z}{\partial z}\hat{e}_z$$

$$\hat{e}_\theta = \hat{e}_z$$

$$\Delta z \downarrow \qquad (J^{-1}) = \begin{pmatrix} \cos\theta & -\rho\operatorname{sen}\theta & 0 \\ \operatorname{sen}\theta & \rho\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(\hat{e}_\rho, \hat{e}_\theta, \hat{e}_z) = (\hat{e}_x, \hat{e}_y, e_z)M$$

cilíndricas ← cartesianas

Relación cilíndricas - cartesianas

$$\begin{cases} \rho = \sqrt{x^2 + y^2} \\ \theta = \operatorname{arctg}\left(\frac{x}{y}\right) \\ z = z \end{cases}$$

$\frac{d}{dy}\sqrt{x^2+y^2} = \frac{(x^2+y^2)^{-1/2}}{2}2x$

$= \frac{1}{\rho}\cdot\rho\cos\theta$

$= \cos\theta$

$\frac{d}{dy}\sqrt{x^2+y^2} = \frac{(x^2+y^2)^{-1/2}}{2}2y$

$= \operatorname{sen}\theta$

$\frac{d}{dx}\operatorname{arctg}\left(\frac{x}{y}\right) = \frac{1}{1+(x/y)^2}\cdot\frac{1}{y}$

$= \frac{y}{y^2+x^2} = \frac{\rho\operatorname{sen}\theta}{\rho^2} = \frac{1}{\rho}\operatorname{sen}\theta$

$\frac{d}{dy}\operatorname{arctg}\left(\frac{x}{y}\right) = \frac{-1}{1+(x/y)^2}\cdot y^{-2}\cdot x$

$= -\frac{1}{\frac{\rho^2}{y^2}}\cdot\frac{1}{y^2}\cdot x = -\frac{1}{\rho^2}\cdot\rho\cos\theta = -\frac{\cos\theta}{\rho}$

• Componentes contravariantes:

cambio de componentes:

$$A'^a = \frac{\partial x'^a}{\partial x^b}A^b \qquad A'^\rho = \frac{\partial x'^\rho}{\partial x^b}A^b$$

$$A'^\rho = \frac{\partial \rho}{\partial x}A^x + \frac{\partial \rho}{\partial y}A^y + \frac{\partial \rho}{\partial z}A^z$$

$$A'^\rho = \cos\theta\, A^x + \operatorname{sen}\theta\, A^y$$

$$A'^\theta = \frac{\partial \theta}{\partial x}A^x + \frac{\partial \theta}{\partial y}A^y + \frac{\partial \theta}{\partial z}A^z$$

$$A'^\theta = \frac{-1}{\sqrt{\rho^2 - x^2}}A^x + \frac{1}{\sqrt{\rho^2 - y^2}}A^\theta$$

$$A'^z = \frac{\partial z}{\partial x}A^x + \frac{\partial z}{\partial y}A^y + \frac{\partial z}{\partial z}A^z$$

$$A'^z = A^z$$

$$A^\rho \rightarrow \qquad M = \begin{pmatrix} \cos\theta & \operatorname{sen}\theta & 0 \\ -\frac{\operatorname{sen}\theta}{\rho} & \frac{\cos\theta}{\rho} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \text{Jacobiano}$$

$$\begin{pmatrix} A^\rho \\ A^\theta \\ A^z \end{pmatrix} = J\begin{pmatrix} A^x \\ A^y \\ A^z \end{pmatrix}$$

Coordenadas cilíndricas ← Coordenadas cartesianas

$$w'_i v'^i = \left(\frac{\partial x^j}{\partial x'^i}\cdot w_j\right)\left(\frac{\partial x'^i}{\partial x^k}V^k\right) = \frac{\partial x^j}{\partial x'^i}\frac{\partial x'^i}{\partial x^k}w_j V^k = \delta^j_k w_j V^k = w_j V^j$$

Producto escalar: $\langle \vec{x}, \vec{y} \rangle = \mathbf{u}\cdot\mathbf{v} = g_{ij}u^i v^j = u_j V^j$

3 si u y V son contravariantes $= (u)^T[g](V)$

si u y V son covariantes $= (u)^T[g]^{-1}(V)$

(contra, cova) (contra, cova)

$$\begin{cases} V_{cov} = g\, V_{contra} \\ V_{contra} = g^{-1} V_{cov} \end{cases}$$

$$A'^a = (J)A^b$$

(contravariante, covariante) ⇒ sin g

$$\Rightarrow w'\cdot A' = (w'_a)^T A'^a = ((J^{-1})^T w_b)^T (J) A^b$$

$$= (w_b)^T J^{-1} J A^b$$

$$= (w_b)^T A^b$$

$$w'\cdot A' = w\cdot A$$

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b) Encuentre la matriz de transformación entre las bases de coordenadas cartesianas y las bases de coordenadas esféricas. **(1p)**

- Componentes Covariantes

$$\hat{e}'_a = \frac{\partial x^b}{\partial x'^a}\hat{e}_b$$

Relación cartesianas - esféricas

$x = r\,\text{sen}\theta\cos\phi$

$y = r\,\text{sen}\theta\,\text{sen}\phi$

$z = r\cos\theta$

$$\rightarrow \hat{e}'_r = \frac{\partial x^b}{\partial r}\hat{e}_b$$

$$\hat{e}'_r = \frac{\partial x}{\partial r}\hat{e}_x + \frac{\partial y}{\partial r}\hat{e}_y + \frac{\partial z}{\partial r}\hat{e}_z$$

$$\hat{e}'_r = \text{sen}\theta\cos\phi\,\hat{e}_x + \text{sen}\theta\,\text{sen}\phi\,\hat{e}_y + \cos\theta\,\hat{e}_z$$

$$\hat{e}'_\theta = \frac{\partial x}{\partial \theta}\hat{e}_x + \frac{\partial y}{\partial \theta}\hat{e}_y + \frac{\partial z}{\partial \theta}\hat{e}_z$$

$$\hat{e}'_r = r\cos\theta\cos\phi\,\hat{e}_x + r\cos\theta\,\text{sen}\phi\,\hat{e}_y - r\,\text{sen}\theta\,\hat{e}_z$$

$$\hat{e}'_\phi = \frac{\partial x}{\partial \phi}\hat{e}_x + \frac{\partial y}{\partial \phi}\hat{e}_y + \frac{\partial z}{\partial \phi}\hat{e}_z$$

$$\hat{e}'_\phi = -r\,\text{sen}\theta\,\text{sen}\phi\,\hat{e}_x + r\,\text{sen}\theta\cos\phi\,\hat{e}_y$$

$$\rightarrow J^{-1} = \begin{pmatrix} \text{sen}\theta\cos\phi & r\cos\theta\cos\phi & -r\,\text{sen}\theta\,\text{sen}\phi \\ \text{sen}\theta\,\text{sen}\phi & r\cos\theta\,\text{sen}\phi & r\,\text{sen}\theta\cos\phi \\ \cos\theta & -r\,\text{sen}\theta & 0 \end{pmatrix} \downarrow \Delta z$$

$$(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi) = (\hat{e}_x, \hat{e}_y, e_z)M$$

esféricas ← cartesianas

c) Encuentre la matriz de transformación entre las bases de coordenadas cilíndricas y las bases de coordenadas esféricas. **(1p)**

- Componentes Covariantes

$$\hat{e}'_a = \frac{\partial x^b}{\partial x'^a}\hat{e}_b$$

Relación cilíndricas - esféricas

$\rho = r\,\text{sen}\theta$

$z = r\cos\theta$

$\phi = \phi$

$$\rightarrow \hat{e}'_r = \frac{\partial x^b}{\partial r}\hat{e}_b \qquad (\rho, \phi, z) \rightarrow (r, \theta, \phi)$$

$$\hat{e}'_r = \frac{\partial \rho}{\partial r}\hat{e}_\rho + \frac{\partial \phi}{\partial r}\hat{e}_\phi + \frac{\partial z}{\partial r}\hat{e}_z$$

$$\hat{e}'_r = \text{sen}\theta\,\hat{e}_\rho + 0\,\hat{e}_\phi + \cos\theta\,\hat{e}_z$$

$$\hat{e}'_\theta = \frac{\partial \rho}{\partial \theta}\hat{e}_\rho + \frac{\partial \phi}{\partial \theta}\hat{e}_\phi + \frac{\partial z}{\partial \theta}\hat{e}_z$$

$$\hat{e}' = r\cos\theta\,\hat{e}_\rho + 0\,\hat{e}_\phi - r\,\text{sen}\theta\,\hat{e}_z$$

$$\hat{e}'_\phi = \frac{\partial \rho}{\partial \phi}\hat{e}_\rho + \frac{\partial \phi}{\partial \phi}\hat{e}_\phi + \frac{\partial z}{\partial \phi}\hat{e}_z$$

$$\hat{e}_\phi = 0\,\hat{e}_\rho + 1\,\hat{e}_z + 0\,\hat{e}_\phi$$

$$\rightarrow J^{-1} = \begin{pmatrix} \text{sen}\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -r\,\text{sen}\theta & 0 \end{pmatrix} \downarrow \Delta z$$

$$(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi) = (\hat{e}_\rho, \hat{e}_\theta, \hat{e}_z)M$$

esféricas ← cilíndricas

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2. (3p) Dado el tensor de levi-civita de cuarto orden $\epsilon^{\alpha\beta\gamma\delta}$ y la relación

$$\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta} = \begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & \delta^\mu_\gamma & \delta^\mu_\delta \\ \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix},$$

a) $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}$

$$\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta} = \delta^\mu_\alpha \begin{vmatrix} \delta^\nu_\beta & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\beta & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\beta & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} - \delta^\mu_\beta \begin{vmatrix} \delta^\nu_\alpha & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} + \delta^\mu_\gamma \begin{vmatrix} \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\delta \end{vmatrix} - \delta^\mu_\delta \begin{vmatrix} \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\gamma \end{vmatrix}$$

① ② ③ ④

① $$\begin{vmatrix} \delta^\nu_\beta & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\beta & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\beta & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} = \delta^\nu_\beta(\delta^\rho_\gamma\delta^\sigma_\delta - \delta^\sigma_\gamma\delta^\rho_\delta) - \delta^\nu_\gamma(\delta^\rho_\beta\delta^\sigma_\delta - \delta^\rho_\delta\cdot\delta^\sigma_\beta) + \delta^\nu_\delta(\delta^\rho_\beta\cdot\delta^\sigma_\gamma - \delta^\rho_\gamma\delta^\sigma_\beta)$$

② $$\begin{vmatrix} \delta^\nu_\alpha & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} = \delta^\nu_\alpha(\delta^\rho_\gamma\delta^\sigma_\delta - \delta^\rho_\delta\delta^\sigma_\gamma) - \delta^\nu_\gamma(\delta^\rho_\alpha\delta^\sigma_\delta - \delta^\sigma_\alpha\delta^\rho_\delta) + \delta^\nu_\delta(\delta^\rho_\alpha\cdot\delta^\sigma_\gamma - \delta^\sigma_\alpha\cdot\delta^\rho_\gamma)$$

③ $$\begin{vmatrix} \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\delta \end{vmatrix} = \delta^\nu_\alpha(\delta^\rho_\beta\cdot\delta^\sigma_\delta - \delta^\sigma_\beta\cdot\delta^\rho_\delta) - \delta^\nu_\beta(\delta^\rho_\alpha\cdot\delta^\sigma_\delta - \delta^\rho_\delta\cdot\delta^\sigma_\alpha) + \delta^\nu_\delta(\delta^\rho_\alpha\cdot\delta^\sigma_\beta - \delta^\rho_\beta\,\delta^\sigma_\alpha)$$

④ $$\begin{vmatrix} \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\gamma \end{vmatrix} = \delta^\nu_\alpha(\delta^\rho_\beta\cdot\delta^\sigma_\gamma - \delta^\sigma_\beta\cdot\delta^\rho_\gamma) - \delta^\nu_\beta(\delta^\rho_\alpha\cdot\delta^\sigma_\gamma - \delta^\rho_\gamma\cdot\delta^\sigma_\alpha) + \delta^\nu_\gamma(\delta^\rho_\alpha\cdot\delta^\sigma_\beta - \delta^\sigma_\alpha\cdot\delta^\rho_\beta)$$

$$\begin{aligned}\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta} = \delta^\mu_\alpha &\Big(\delta^\nu_\beta(\delta^\rho_\gamma\delta^\sigma_\delta - \delta^\sigma_\gamma\delta^\rho_\delta) - \delta^\nu_\gamma(\delta^\rho_\beta\delta^\sigma_\delta - \delta^\rho_\delta\cdot\delta^\sigma_\beta) + \delta^\nu_\delta(\delta^\rho_\beta\cdot\delta^\sigma_\gamma - \delta^\rho_\gamma\delta^\sigma_\beta)\Big) \\ - \delta^\mu_\beta &\Big(\delta^\nu_\alpha(\delta^\rho_\gamma\delta^\sigma_\delta - \delta^\rho_\delta\delta^\sigma_\gamma) - \delta^\nu_\gamma(\delta^\rho_\alpha\delta^\sigma_\delta - \delta^\sigma_\alpha\delta^\rho_\delta) + \delta^\nu_\delta(\delta^\rho_\alpha\cdot\delta^\sigma_\gamma - \delta^\sigma_\alpha\cdot\delta^\rho_\gamma)\Big) \\ + \delta^\mu_\gamma &\Big(\delta^\nu_\alpha(\delta^\rho_\beta\cdot\delta^\sigma_\delta - \delta^\sigma_\beta\cdot\delta^\rho_\delta) - \delta^\nu_\beta(\delta^\rho_\alpha\cdot\delta^\sigma_\delta - \delta^\rho_\delta\cdot\delta^\sigma_\alpha) + \delta^\nu_\delta(\delta^\rho_\alpha\cdot\delta^\sigma_\beta - \delta^\sigma_\alpha\cdot\delta^\rho_\beta)\Big) \\ - \delta^\mu_\delta &\Big(\delta^\nu_\alpha(\delta^\rho_\beta\cdot\delta^\sigma_\gamma - \delta^\sigma_\beta\cdot\delta^\rho_\gamma) - \delta^\nu_\beta(\delta^\rho_\alpha\cdot\delta^\sigma_\gamma - \delta^\rho_\gamma\cdot\delta^\sigma_\alpha) + \delta^\nu_\gamma(\delta^\rho_\alpha\cdot\delta^\sigma_\beta - \delta^\sigma_\alpha\cdot\delta^\rho_\beta)\Big)\end{aligned}$$

b) $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\sigma}$

$\delta=\sigma$

$$\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\sigma} = \begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & \delta^\mu_\gamma & \cancel{\delta^\mu_\delta}^{\,0} \\ \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma & \cancel{\delta^\nu_\delta}^{\,0} \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma & \cancel{\delta^\rho_\delta}^{\,0} \\ \cancel{\delta^\sigma_\alpha}_{\,0} & \cancel{\delta^\sigma_\beta}_{\,0} & \cancel{\delta^\sigma_\gamma}_{\,0} & \underline{\delta^\sigma_\delta}_{\,1} \end{vmatrix} = -1 \begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & \delta^\mu_\gamma \\ \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma \end{vmatrix} = -\Big(\delta^\mu_\alpha(\delta^\nu_\beta\cdot\delta^\rho_\gamma - \delta^\rho_\beta\cdot\delta^\nu_\gamma) - \delta^\mu_\beta(\delta^\nu_\alpha\cdot\delta^\rho_\gamma - \delta^\rho_\alpha\cdot\delta^\nu_\gamma) + \delta^\mu_\gamma(\delta^\nu_\alpha\cdot\delta^\rho_\beta - \delta^\rho_\alpha\cdot\delta^\nu_\beta)\Big)$$

c) $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\rho\sigma}$

$\delta=\sigma$
$\gamma=\rho$

$$\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\rho\sigma} = n!\begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & \delta^\mu_\gamma & \delta^\mu_\delta \\ \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} = -2!\begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & 0 \\ \delta^\nu_\alpha & \delta^\nu_\beta & 0 \\ 0 & 0 & 1 \end{vmatrix} = -2(\delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha) = 2(\delta^\mu_\beta\delta^\nu_\alpha - \delta^\mu_\alpha\delta^\nu_\beta)$$

$\sum \epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\rho\sigma} =$

$2!\,\epsilon^{\mu\nu}\epsilon_{\alpha\beta}$
↑
permutaciones

d) $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\nu\rho\sigma}$

$$\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\nu\rho\sigma} = n!\begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & \delta^\mu_\gamma & \delta^\mu_\delta \\ \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} = 3!\,\delta^\mu_\alpha = 6\,\delta^\mu_\alpha$$

$\nu=\beta$
$\delta=\sigma$
$\gamma=\rho$

e) $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\sigma}$

$$\epsilon^{\alpha\nu\rho\sigma}\epsilon_{\alpha\nu\rho\sigma} = n!\begin{vmatrix} \delta^\mu_\alpha & \delta^\mu_\beta & \delta^\mu_\gamma & \delta^\mu_\delta \\ \delta^\nu_\alpha & \delta^\nu_\beta & \delta^\nu_\gamma & \delta^\nu_\delta \\ \delta^\rho_\alpha & \delta^\rho_\beta & \delta^\rho_\gamma & \delta^\rho_\delta \\ \delta^\sigma_\alpha & \delta^\sigma_\beta & \delta^\sigma_\gamma & \delta^\sigma_\delta \end{vmatrix} = 4!\,\delta^\alpha_\alpha = 24$$

$\alpha=\alpha$
$\nu=\beta$
$\delta=\sigma$
$\gamma=\rho$

3. Para tensores antisimétricos de 3er y 4to rango, mostrar que:

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a) $T_{[\alpha\beta\gamma]} = \frac{1}{3!}(T_{\alpha[\beta\gamma]} - T_{\beta[\alpha\gamma]} + T_{\gamma[\alpha\beta]})$ **(1.5p)**

La antisimetrización completa de un tensor de rango n,

$$T_{[a_1,a_2,\ldots,a_n]} = \frac{1}{n!}\sum_\pi \text{sign}(\pi)\, T_{a_{\pi(1)}a_{\pi(2)}\ldots a_{\pi(n)}}$$

Demostración:

$$T_{[\alpha\beta\gamma]} = \frac{1}{6}\left(T_{\alpha\beta\sigma} - T_{\alpha\sigma\beta} - T_{\beta\alpha\sigma} + T_{\beta\sigma\alpha} + T_{\sigma\alpha\beta} - T_{\sigma\beta\alpha}\right)$$

antisimétrico

$T_{\alpha[\beta\sigma]} = \frac{1}{2}(T_{\alpha\beta\sigma} - T_{\alpha\sigma\beta})$
$T_{\beta[\alpha\sigma]} = \frac{1}{2}(T_{\beta\alpha\sigma} - T_{\beta\sigma\alpha})$
$T_{\gamma[\alpha\beta]} = \frac{1}{2}(T_{\gamma\alpha\beta} - T_{\gamma\beta\alpha})$

sustituimos

$$T_{[\alpha\beta\gamma]} = \frac{1}{3}\left(\frac{1}{2}(T_{\alpha\beta\sigma} - T_{\alpha\sigma\beta}) - \frac{1}{2}(T_{\beta\alpha\sigma} - T_{\beta\sigma\alpha}) + \frac{1}{2}(T_{\sigma\alpha\beta} - T_{\sigma\beta\alpha})\right)$$

permutaciones

$$T_{[\alpha\beta\gamma]} = \frac{1}{3!}(T_{\alpha[\beta\gamma]} - T_{\beta[\alpha\gamma]} + T_{\gamma[\alpha\beta]})$$

$$b)\ T_{[\alpha\beta\gamma\delta]} = \frac{1}{4!}(T_{\alpha[\beta\gamma\delta]} - T_{\beta[\alpha\gamma\delta]} + T_{\gamma[\alpha\beta\delta]} - T_{\delta[\alpha\beta\gamma]})\ \textbf{(1.5p)}$$

Demostración:

sustituimos en a)

$$T_{[\alpha\beta\gamma\delta]} = \frac{1}{16}\Big(T^{1234}_{\alpha\beta\gamma\delta} - T^{1324}_{\alpha\gamma\beta\delta} + T^{1342}_{\alpha\gamma\delta\beta} - T^{1432}_{\alpha\delta\gamma\beta} + T^{1423}_{\alpha\delta\beta\gamma} - T^{1243}_{\alpha\beta\delta\gamma}$$
$$- T^{2134}_{\beta\alpha\gamma\delta} + T^{2314}_{\beta\gamma\alpha\delta} - T^{2341}_{\beta\gamma\delta\alpha} + T^{2431}_{\beta\delta\gamma\alpha} - T^{2413}_{\beta\delta\alpha\gamma} + T^{2143}_{\beta\gamma\delta\alpha}$$
$$T^{3124}_{\gamma\alpha\beta\delta} - T^{3214}_{\gamma\beta\alpha\delta} + T^{3241}_{\gamma\beta\delta\alpha} - T^{3421}_{\gamma\delta\beta\alpha} + T^{3412}_{\gamma\delta\alpha\beta} - T^{3142}_{\gamma\alpha\delta\beta}$$
$$- T^{4312}_{\delta\gamma\alpha\beta} + T^{4132}_{\delta\alpha\gamma\beta} - T^{4123}_{\delta\alpha\beta\gamma} + T^{4213}_{\delta\beta\alpha\gamma} - T^{4231}_{\delta\beta\gamma\alpha} + T^{4321}_{\delta\gamma\beta\alpha}\Big)$$

antisimétrico

$$T_{\alpha[\beta\gamma\delta]} = \tfrac{1}{6}(T^{1234}_{\alpha\beta\gamma\delta} - T^{1324}_{\alpha\gamma\beta\delta} + T^{1342}_{\alpha\gamma\delta\beta} - T^{1432}_{\alpha\delta\gamma\beta} + T^{1423}_{\alpha\delta\beta\gamma} - T^{1243}_{\alpha\beta\delta\gamma})$$
$$T_{\beta[\alpha\gamma\delta]} = \tfrac{1}{6}(T^{1234}_{\beta\alpha\gamma\delta} - T^{1324}_{\beta\gamma\alpha\delta} + T^{1342}_{\beta\gamma\delta\alpha} - T^{1432}_{\beta\delta\gamma\alpha} + T^{1423}_{\beta\delta\alpha\gamma} - T^{1243}_{\beta\gamma\delta\alpha})$$
$$T_{\gamma[\alpha\beta\delta]} = \tfrac{1}{6}(T^{1234}_{\gamma\alpha\beta\delta} - T^{1324}_{\gamma\beta\alpha\delta} + T^{1342}_{\gamma\beta\delta\alpha} - T^{1432}_{\gamma\delta\beta\alpha} + T^{1423}_{\gamma\delta\alpha\beta} - T^{1243}_{\gamma\alpha\delta\beta})$$
$$T_{\delta[\alpha\beta\gamma]} = \tfrac{1}{6}(T^{1234}_{\delta\gamma\alpha\beta} - T^{1324}_{\delta\alpha\gamma\beta} + T^{1342}_{\delta\alpha\beta\gamma} - T^{1432}_{\delta\beta\gamma\alpha} + T^{1423}_{\delta\beta\gamma\alpha} - T^{1243}_{\delta\gamma\beta\alpha})$$

$$T_{[\alpha\beta\gamma\delta]} = \frac{1}{4}\Big(\tfrac{1}{6}T_{\alpha[\beta\gamma\delta]} - \tfrac{1}{6}T_{\beta[\alpha\gamma\delta]} + \tfrac{1}{6}T_{\gamma[\alpha\beta\delta]} - \tfrac{1}{6}T_{\delta[\alpha\beta\gamma]}\Big)$$

$$T_{[\alpha\beta\gamma\delta]} = \frac{1}{24}\Big(T_{\alpha[\beta\gamma\delta]} - T_{\beta[\alpha\gamma\delta]} + T_{\gamma[\alpha\beta\delta]} - T_{\delta[\alpha\beta\gamma]}\Big)$$

$4! = 4.3.2$

4. Sea A^μ un cuadrivector. Encuentre la regla de transformación de las componentes $\partial_\nu A^\mu$ de la gradiente de A^μ y demuestre que estas se transforman como tensor únicamente ante transformaciones de coordenadas lineales. **(1p)**

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$$D[f] = D^\mu \frac{\partial}{\partial x^\mu} f$$

$f(x(\lambda))$ (vec, esc)

$$= \frac{\partial x^\mu}{\partial \lambda}\frac{\partial}{\partial x^\mu} f$$

tomando a

D^μ ∧ $\hat{e}_\mu = \frac{\partial}{\partial x^\mu}$

vector contravariante — vector covariante

cumple bajo transformaciones

$$D[f] = D^\alpha \frac{\partial}{\partial y^\alpha} f \quad \dots(1)$$

$f(Y(\lambda))$

Pero:

$Y = f(x)$

$f = f(y(x(\lambda)))$

por regla de la cadena:

$$\Rightarrow D[f] = \frac{\partial f}{\partial y^\alpha}\cdot\frac{\partial y^\alpha}{\partial x^\mu}\cdot\frac{\partial x^\mu}{\partial \lambda}$$

$$D[f] = D^\mu \cdot \frac{\partial y^\alpha}{\partial x^\mu}\cdot\frac{\partial}{\partial y^\alpha} f$$

$$D[f] = D^\alpha \frac{\partial}{\partial y^\alpha} f$$

Para una transformación Contravariante

$$V^\alpha = \frac{\partial y^\alpha}{\partial x^\mu} V^\mu$$

$$D^\alpha = D^\mu \frac{\partial y^\alpha}{\partial x^\mu}$$

Para una transformación Covariante

$$W_\alpha = \frac{\partial x^\mu}{\partial y^\alpha} W_\mu$$

$$D^\alpha W_\alpha = D^\mu \frac{\partial y^\alpha}{\partial x^\mu} W_\alpha = D^\mu W_\mu$$

$$W_\alpha = \frac{\partial x^\mu}{\partial y^\alpha} W_\mu$$

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$$A'^{\alpha}(x) = \frac{\partial x'^{\alpha}}{\partial x^{\mu}} A^{\mu} \quad \dots (I)$$

Coordenadas originales: x^{μ} $\quad x = (x^1, \dots, x^{\mu})$

Coordenadas nuevas: $x'^{\alpha} = x'^{\alpha}(x)$

- $\partial'_{\beta} A'^{\alpha} = \frac{\partial A'^{\alpha}}{\partial x'^{\beta}}$

$$= \frac{\partial A'^{\mu}}{\partial x^{\nu}} \cdot \frac{\partial x^{\nu}}{\partial x'^{\beta}}$$

$$\partial'_{\beta} A'^{\alpha} = \frac{\partial x^{\nu}}{\partial x'^{\beta}} \cdot \frac{\partial}{\partial x^{\nu}} (A'^{\alpha}) \rightarrow (I)$$

$$\partial'_{\beta} A'^{\alpha} = \frac{\partial x^{\nu}}{\partial x'^{\beta}} \cdot \frac{\partial}{\partial x^{\nu}} \left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}} A^{\mu} \right]$$

$$\partial'_{\beta} A'^{\alpha} = \frac{\partial x^{\nu}}{\partial x'^{\beta}} \cdot \left[\frac{\partial^2 x'^{\alpha}}{\partial x^{\nu} \partial x^{\mu}} A^{\mu} + \frac{\partial x'^{\alpha}}{\partial x^{\mu}} \frac{\partial A^{\mu}}{\partial x^{\nu}} \right]$$

$$\partial'_{\beta} A'^{\alpha} = \frac{\partial x^{\nu}}{\partial x'^{\beta}} \frac{\partial x'^{\alpha}}{\partial x^{\mu}} \frac{\partial A^{\mu}}{\partial x^{\nu}} + \frac{\partial x^{\nu}}{\partial x'^{\beta}} \frac{\partial^2 x'^{\alpha}}{\partial x^{\nu} \partial x^{\mu}} A^{\mu}$$

$$\partial'_{\beta} A'^{\alpha} = \frac{\partial x^{\nu}}{\partial x'^{\beta}} \frac{\partial x'^{\alpha}}{\partial x^{\mu}} \partial_{\nu} A^{\mu} + \frac{\partial x^{\nu}}{\partial x'^{\beta}} \frac{\partial^2 x'^{\alpha}}{\partial x^{\nu} \partial x^{\mu}} A^{\mu}$$

Pero tenemos este extra

y si la transformación fuera lineal

$$x'^{\alpha} = L^{\alpha}{}_{\mu} x^{\mu} + a^{\alpha}$$

$$\frac{\partial^2 x'^{\alpha}}{\partial x^{\nu} \partial x^{\mu}} = \frac{\partial}{\partial x^{\nu}} \left(\frac{\partial x'^{\alpha}}{\partial x^{\mu}} \right) = \frac{\partial}{\partial x^{\nu}} L^{\alpha}{}_{\mu} = 0$$

Si fuera un tensor (1,1):

$$T'^{\alpha}{}_{\beta} = \frac{\partial x^{\nu}}{\partial x'^{\beta}} \frac{\partial x'^{\alpha}}{\partial x^{\mu}} T^{\mu}{}_{\nu}$$

$\Rightarrow$ Las componentes se transforman como un tensor solo bajo transformaciones lineales.

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5. Considere las coordenadas polares en un espacio bidimensional plano. Las ecuaciones de transformación de coordenadas entre las coordenadas polares (r, θ), a las cuales consideraremos como coordenadas primadas, y las cartesianas (x, y) son

$$x = r\cos\theta \qquad r = \sqrt{x^2 + y^2}$$
$$y = r\sin\theta \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Considere también la función escalar $\phi = bxy = br^2\cos(\theta)\sin(\theta)$, donde b es una constante arbitraria.

a) Calcule las cuatro derivadas parciales $\dfrac{\partial x^\mu}{\partial x^{\nu'}}$ para las transformaciones mostradas. **(1p)**

$\hat{e}_b$ (coordenadas cartesianas)

$\hat{e}'_a$ (coordenadas polares)

Cambio de base:

$$\hat{e}'_a = \frac{\partial x^b}{\partial x'^a}\hat{e}_b$$

$$\Rightarrow \hat{e}_r = \frac{\partial x}{\partial r}\hat{e}_x + \frac{\partial y}{\partial r}\hat{e}_y \qquad \hat{e}_\theta = \frac{\partial x}{\partial \theta}\hat{e}_x + \frac{\partial y}{\partial \theta}\hat{e}_y$$

$$\hat{e}_r = \cos\theta\,\hat{e}_x + \operatorname{sen}\theta\,\hat{e}_y \qquad \hat{e}_\theta = -r\operatorname{sen}\theta\,\hat{e}_x + r\cos\theta\,\hat{e}_y$$

$X = r\cos\theta$

$Y = r\operatorname{sen}\theta$

$$\frac{\partial x^1}{\partial x'^1} = \frac{\partial x}{\partial r} = \cos\theta \qquad \frac{\partial x^2}{\partial x'^1} = \frac{\partial Y}{\partial r} = \operatorname{sen}\theta$$

$$\frac{\partial x^1}{\partial x^{2'}} = \frac{\partial x}{\partial \theta} = -r\operatorname{sen}\theta \qquad \frac{\partial x^2}{\partial x^{2'}} = \frac{\partial Y}{\partial \theta} = r\cos\theta$$

b) Use la definición básica de la gradiente para calcular las componentes $\partial_\mu\phi$ de la gradiente de ϕ en ambos sistemas coordenados. **(1p)**

$$\phi = bxy = br^2\cos(\theta)\operatorname{sen}(\theta) = \frac{br^2}{2}\operatorname{sen}(2\theta)$$

$$\nabla\phi = \partial_\mu\phi$$

En coordenadas cartesianas $(x^1 = x, x^2 = y)$:

$\partial_\mu\phi$:

$$\left.\begin{array}{l} x^1 = x \rightarrow \dfrac{\partial}{\partial x}\phi = by \\ x^2 = y \rightarrow \dfrac{\partial}{\partial y}\phi = bx \end{array}\right\} \partial_\mu\phi = (by, bx)$$

En coordenadas polares $(x^1 = r, x^2 = \theta)$:

$\partial_\mu\phi$:

$$\left.\begin{array}{l} x^1 = r \rightarrow \dfrac{\partial}{\partial r}\phi = br\operatorname{sen}2\theta \\ x^2 = \theta \rightarrow \dfrac{\partial}{\partial \theta}\phi = br^2\cos 2\theta \end{array}\right\} \partial_\mu\phi = (br\operatorname{sen}2\theta, br^2\cos 2\theta)$$

$\partial_\mu \phi$

c) Demuestre que si se toma las componentes de la gradiente de ϕ en el sistema de coordenado cartesiano y se las transforma utilizando la regla de transformación de tensor covariante de rango 1, se obtienen las componentes de la gradiente en el sistema coordenado polar. **(1p)**

$$w'_\alpha = \frac{\partial x^\mu}{\partial x'^\alpha} w_\mu$$

Polares ← Cartesianos

- $w'_\alpha = \dfrac{\partial \phi}{\partial x'^\alpha}$
- $w_\mu = \dfrac{\partial \phi}{\partial x^\mu}$

$$\frac{\partial \phi}{\partial x'^\alpha} = \frac{\partial x^\mu}{\partial x'^\alpha} \frac{\partial \phi}{\partial x^\mu}$$

$$\frac{\partial \phi}{\partial r} = \frac{\partial x}{\partial r}\cdot\frac{\partial \phi}{\partial x} + \frac{\partial y}{\partial r}\cdot\frac{\partial \phi}{\partial y}$$

$$\frac{\partial \phi}{\partial \theta} = \frac{\partial x}{\partial \theta}\cdot\frac{\partial \phi}{\partial x} + \frac{\partial y}{\partial \theta}\cdot\frac{\partial \phi}{\partial y}$$

$$br\,\mathrm{sen}\,2\theta = \cos\theta \cdot by + \mathrm{sen}\theta\, bx$$

$$br\,\mathrm{sen}\,2\theta = br(\cos\theta\,\mathrm{sen}\theta + \mathrm{sen}\theta\cos\theta)$$

$$br\,\mathrm{sen}\,2\theta \equiv br\,\mathrm{sen}\,2\theta$$

$$br^2\cos 2\theta = -r\,\mathrm{sen}\theta \cdot by + r\cos\theta\, bx$$

$$= r^2 b(-\mathrm{sen}\theta\cdot\mathrm{sen}\theta + \cos\theta\cdot\cos\theta)$$

$$br^2\cos 2\theta \equiv br^2\cos 2\theta$$

usando:

$$\frac{\partial x}{\partial r} = \cos\theta \;,\; \frac{\partial y}{\partial r} = \mathrm{sen}\theta$$

$$\frac{\partial x}{\partial \theta} = -r\,\mathrm{sen}\theta \;,\; \frac{\partial y}{\partial \theta} = r\cos\theta$$

$\rightsquigarrow \dfrac{\partial}{\partial x}\phi = by$

$\rightsquigarrow \dfrac{\partial}{\partial y}\phi = bx$

$\rightsquigarrow \dfrac{\partial}{\partial r}\phi = br\,\mathrm{sen}\,2\theta$

$\rightsquigarrow \dfrac{\partial}{\partial \theta}\phi = br^2\cos 2\theta$

$x = r\cos\theta$

$y = r\,\mathrm{sen}\,\theta$

6. Sean A^α, $B^{\alpha\beta\rho}$ y $C^{\alpha\beta\rho\sigma}$ las componentes contravariantes de tres tensores en cierto sistema coordenado. Demuestre que

$$T^{\alpha\beta}{}_{\rho\gamma\sigma} = \overbrace{A^\alpha B_{\rho\gamma\sigma} A^\beta}^{\text{Tensor (2,3)}} - \overbrace{B^{\alpha\beta\omega} C_{\omega\rho\gamma\sigma}}^{\text{Tensor (2,3}}$$

$B^{\alpha\beta}\, C_{\rho\sigma}$

representa a las componentes de un tensor. **(1p)**

usaremos: $\alpha', \beta', \rho', \gamma', \sigma'$ como coordenadas después de la transformación.

$$A^{\alpha'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha} A^\alpha$$

$$B_{\rho'\gamma'\sigma'} = \frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} B_{\rho\gamma\sigma}$$

$$A^{\beta'} = \frac{\partial x^{\beta'}}{\partial x^\beta} A^\beta$$

$$A^{\alpha'} B_{\rho'\gamma'\sigma'} A^{\beta'} = \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} A^\alpha\right)\left(\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} B_{\rho\gamma\sigma}\right)\left(\frac{\partial x^{\beta'}}{\partial x^\beta} A^\beta\right)$$

$$B^{\alpha'\beta'\omega'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha}\frac{\partial x^{\beta'}}{\partial x^\beta}\frac{\partial x^{\omega'}}{\partial x^\omega} B^{\alpha\beta\omega}$$

$$C_{\omega'\rho'\gamma'\sigma'} = \frac{\partial x^\omega}{\partial x^{\omega'}}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} C_{\omega\rho\gamma\sigma}$$

$$B^{\alpha'\beta'\omega'} C_{\omega'\rho'\gamma'\sigma'} = \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha}\frac{\partial x^{\beta'}}{\partial x^\beta}\frac{\partial x^{\omega'}}{\partial x^\omega} B^{\alpha\beta\omega}\right)\left(\frac{\partial x^\omega}{\partial x^{\omega'}}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} C_{\omega\rho\gamma\sigma}\right)$$

$$= \frac{\partial x^{\alpha'}}{\partial x^\alpha}\frac{\partial x^{\beta'}}{\partial x^\beta}\underbrace{\frac{\partial x^{\omega'}}{\partial x^\omega}\frac{\partial x^\omega}{\partial x^{\omega'}}}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} B^{\alpha\beta\omega} C_{\omega\rho\gamma\sigma}$$

$$\sum_\omega \frac{\partial x^{\omega'}}{\partial x^\omega}\frac{\partial x^\omega}{\partial x^{\omega'}} = \sum_\omega \delta^{\omega'}_\omega \delta^\omega_{\omega'} = \delta^{\omega'}_{\omega'} = 1$$

ω' es fijo (no se repite)

$$B^{\alpha'\beta'\omega'} C_{\omega'\rho'\gamma'\sigma'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha}\frac{\partial x^{\beta'}}{\partial x^\beta}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} B^{\alpha\beta\omega} C_{\omega\rho\gamma\sigma}$$

$$Z^{\alpha'\beta'}{}_{\rho'\gamma'\sigma'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha}\frac{\partial x^{\beta'}}{\partial x^\beta}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} Z^{\alpha\beta}{}_{\rho\gamma\sigma}$$

↝ transforma como un tensor (2,3)

Tensores del mismo tipo (2,3)

$$T^{\alpha'\beta'}{}_{\rho'\gamma'\sigma'} = A^{\alpha'} B_{\rho'\gamma'\sigma'} A^{\beta'} - B^{\alpha'\beta'\omega'} C_{\omega'\rho'\gamma'\sigma'}$$

$\Rightarrow$ La resta también será un tensor (2,3)

$$T^{\alpha'\beta'}{}_{\rho'\gamma'\sigma'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha}\frac{\partial x^{\beta'}}{\partial x^\beta}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\gamma}{\partial x^{\gamma'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}} T^{\alpha\beta}{}_{\rho\gamma\sigma}$$

□

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7. Dado un tensor de componentes $T^{\alpha\beta}{}_{\gamma\rho}$:

a) Encuentre una expresión para $T^{\alpha}{}_{\beta\gamma\rho}$ y demuestre que representa las componentes de un tensor. **(1.5p)**

Tensor $T^{\alpha\beta}{}_{\gamma\rho}$ (rango (2,2)), número total de componentes: n^4

hay $n\times n\times n$ matrices por cada β, en total hay $n\times n\times n\times n = n^4$.

$$T^{\alpha\beta}{}_{\gamma\rho} = \left(\begin{pmatrix} T^{11}_{11} & T^{12}_{11} & \cdots & T^{1n}_{11} \\ T^{21}_{11} & T^{22}_{11} & \cdots & T^{2n}_{11} \\ \vdots & \vdots & \ddots & \vdots \\ T^{n1}_{11} & T^{n2}_{11} & \cdots & T^{nn}_{11} \end{pmatrix}_{\gamma=1} \cdots \gamma=n \right)_{\beta=1}$$

$\beta=1$

$$\left(\begin{pmatrix} T^{11}_{12} & T^{12}_{12} & \cdots & T^{1n}_{12} \\ T^{21}_{12} & T^{22}_{12} & \cdots & T^{2n}_{12} \\ \vdots & \vdots & \ddots & \vdots \\ T^{n1}_{12} & T^{n2}_{12} & \cdots & T^{nn}_{12} \end{pmatrix}_{\gamma=1} \cdots \gamma=n \right)_{\beta=2}$$

$\beta=2$

$$\cdots \left(\begin{pmatrix} T^{11}_{1\rho} & T^{12}_{1\rho} & \cdots & T^{1n}_{1\rho} \\ T^{21}_{1\rho} & T^{22}_{\gamma\rho} & \cdots & T^{2n}_{\gamma\rho} \\ \vdots & \vdots & \ddots & \vdots \\ T^{n1}_{\gamma\rho} & T^{n2}_{\gamma\rho} & \cdots & T^{nn}_{\gamma\rho} \end{pmatrix}_{\gamma=1} \cdots \gamma=n \right)_{\beta=n}$$

$\beta=n$

$$T^{\alpha}{}_{\beta\gamma\rho} = g_{\beta\mu} T^{\alpha\mu}{}_{\gamma\rho}$$

$x^{\mu} \to x^{\mu'}$:

$$g_{\beta'\mu'} = \frac{\partial x^{\nu}}{\partial x^{\beta'}} \frac{\partial x^{\sigma}}{\partial x^{\mu'}} g_{\nu\sigma}$$

$$T^{\alpha'\mu'}{}_{\gamma'\rho'} = \frac{\partial x^{\alpha'}}{\partial x^{\alpha}} \frac{\partial x^{\mu'}}{\partial x^{\mu}} \frac{\partial x^{\gamma}}{\partial x^{\gamma'}} \frac{\partial x^{\rho}}{\partial x^{\rho'}} T^{\alpha\mu}{}_{\gamma\rho}$$

$$T^{\alpha'}{}_{\beta'\gamma'\rho'} = \frac{\partial x^{\nu}}{\partial x^{\beta'}} \frac{\partial x^{\sigma}}{\partial x^{\mu'}} g_{\nu\sigma} \frac{\partial x^{\alpha'}}{\partial x^{\alpha}} \frac{\partial x^{\mu'}}{\partial x^{\mu}} \frac{\partial x^{\gamma}}{\partial x^{\gamma'}} \frac{\partial x^{\rho}}{\partial x^{\rho'}} T^{\alpha\mu}{}_{\gamma\rho}$$

$$\sum_{\mu'} \frac{\partial x^{\sigma}}{\partial x^{\mu'}} \frac{\partial x^{\mu'}}{\partial x^{\mu}} = \delta^{\sigma}_{\mu}$$

$$\delta^{\sigma}_{\mu} g_{\nu\sigma} T^{\alpha\mu}{}_{\gamma\rho} = g_{\nu\mu} T^{\alpha\mu}{}_{\gamma\rho} = T^{\alpha}{}_{\nu\gamma\rho}$$

$$T^{\alpha'}{}_{\beta'\gamma'\rho'} = \frac{\partial x^{\alpha'}}{\partial x^{\alpha}} \frac{\partial x^{\nu}}{\partial x^{\beta'}} \frac{\partial x^{\gamma}}{\partial x^{\gamma'}} \frac{\partial x^{\rho}}{\partial x^{\rho'}} T^{\alpha}{}_{\nu\gamma\rho}$$

Tensor $T^{\alpha}{}_{\beta\gamma\rho}$ (rango (1,3)), número total de componentes: n^4

$$A_{\mu} = g_{\mu\nu} A^{\nu} \quad \cdots (I)$$

$$g_{\mu'\nu'} = \frac{\partial x^{\alpha}}{\partial x^{\mu'}} \frac{\partial x^{\beta}}{\partial x^{\nu'}} g_{\alpha\beta}$$

$$g_{\mu'\nu'} A^{\nu'} = \left(\frac{\partial x^{\alpha}}{\partial x^{\mu'}} \frac{\partial x^{\beta}}{\partial x^{\nu'}} g_{\alpha\beta}\right)\left(\frac{\partial x^{\nu'}}{\partial x^{\nu}} A^{\nu}\right)$$

$$A_{\mu'} = \frac{\partial x^{\beta}}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^{\nu}} = \delta^{\beta}_{\nu} \rightarrow g_{\alpha\rho} = g_{\alpha\nu}$$

cambio de índice mudo:

$$\boxed{g_{\mu'\nu'} A^{\nu'} = \frac{\partial x^{\alpha}}{\partial x^{\mu'}} g_{\alpha\nu} A^{\nu} = \frac{\partial x^{\mu}}{\partial x^{\mu'}} A_{\mu}}$$

($g_{\mu'\nu'}A^{\nu'} = A_{\mu'}$, $g_{\alpha\nu}A^{\nu} = A_{\alpha}$)

A_{μ} se transforma como covariante

b) Encuentre una expresión para $T^{\alpha\beta}{}_{\gamma}{}^{\rho}$ y demuestre que representa las componentes de un tensor. **(1.5p)**

$$T^{\alpha\beta\ \rho}_{\ \ \gamma} = g^{\rho\mu}\, T^{\alpha\mu}_{\ \ \gamma\rho}$$

$x^{\mu} \to x^{\mu'}$: $g^{\rho'\mu'} = \dfrac{\partial x^{\rho'}}{\partial x^{\nu}}\dfrac{\partial x^{\mu'}}{\partial x^{\sigma}}\, g^{\nu\sigma}$ Por definición de tensor contra.

$$T^{\alpha'\mu'}{}_{\gamma'\rho'} = \frac{\partial x^{\alpha'}}{\partial x^{\alpha}}\frac{\partial x^{\mu'}}{\partial x^{\mu}}\frac{\partial x^{\gamma}}{\partial x^{\gamma'}}\frac{\partial x^{\rho}}{\partial x^{\rho'}}\, T^{\alpha\mu}{}_{\gamma\rho}$$

$$T^{\alpha'\beta'\ \rho'}_{\ \ \ \gamma'} = \frac{\partial x^{\rho'}}{\partial x^{\nu}}\frac{\partial x^{\mu'}}{\partial x^{\sigma}}\, g^{\nu\sigma}\, \frac{\partial x^{\alpha'}}{\partial x^{\alpha}}\frac{\partial x^{\mu'}}{\partial x^{\mu}}\frac{\partial x^{\gamma}}{\partial x^{\gamma'}}\frac{\partial x^{\rho}}{\partial x^{\rho'}}\, T^{\alpha\mu}{}_{\gamma\rho}$$

$$\sum_{\rho'} \frac{\partial x^{\rho'}}{\partial x^{\nu}}\frac{\partial x^{\rho}}{\partial x^{\rho'}} = \delta^{\rho}_{\nu} \longrightarrow \delta^{\rho}_{\nu}\, g^{\nu\sigma}\, T^{\alpha\mu}{}_{\gamma\rho} = g^{\rho\sigma}\, T^{\alpha\mu}{}_{\gamma\rho} = T^{\alpha\mu}{}_{\gamma}{}^{\sigma}$$

$$T^{\alpha'}{}_{\beta'\gamma'\rho'} = \frac{\partial x^{\alpha'}}{\partial x^{\alpha}}\frac{\partial x^{\nu}}{\partial x^{\beta'}}\frac{\partial x^{\gamma}}{\partial x^{\gamma'}}\frac{\partial x^{\rho}}{\partial x^{\rho'}}\, T^{\alpha}{}_{\nu\gamma\rho}$$

Tensor $T^{\alpha}{}_{\beta\gamma\rho}$, número total de componentes: n^4

(rango (1,3))

hay $n \times n \times n$ matrices por cada ρ,
en total hay $n \times n \times n \times n = n^4$.

$$T^{\alpha}{}_{\beta\gamma\rho} = \underset{\rho=1}{\begin{pmatrix} T^{1}_{111} & T^{1}_{211} & \cdots & T^{1}_{n11} & T^{1}_{n21} & \cdots & T^{1}_{nn1} & T^{1}_{nn2} & \cdots & T^{1}_{nnn} \\ T^{2}_{111} & T^{2}_{211} & \cdots & T^{2}_{n11} & T^{2}_{n21} & \cdots & T^{2}_{nn1} & T^{2}_{nn2} & \cdots & T^{2}_{nnn} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ T^{n}_{111} & T^{n}_{211} & \cdots & T^{n}_{n11} & T^{n}_{n21} & \cdots & T^{n}_{nn1} & T^{n}_{nn2} & \cdots & T^{n}_{nnn} \end{pmatrix}}_{\gamma=1} \cdots \gamma = n$$

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8. Demuestre que $\epsilon^{\alpha\beta\gamma\delta}$ se transforma como una densidad tensorial y que $\xi^{\alpha\beta\gamma\delta} = \sqrt{-g}\epsilon^{\alpha\beta\gamma\delta}$ se transforma como un tensor, donde $\epsilon^{\alpha\beta\gamma\delta}$ representa el símbolo de levi-civita y g el determinante del tensor métrico. **(3p)**

Para cantidades que dependen del volumen

• Una densidad tensorial $T_{\mu\nu\rho\sigma}$ de peso w se transforma bajo un cambio de coordenadas como:

$x^\mu \to x'^\mu$,

$$(I)\ldots T'_{\mu\nu\rho\sigma} = \left|\frac{\partial x'}{\partial x}\right|^w \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma} T_{\alpha\beta\gamma\delta}$$

$\left|\frac{\partial x'}{\partial x}\right|$ Valor absoluto

$\frac{\partial x'}{|\partial x|}$ es el valor absoluto del determinante de la matriz Jacobiana

$J = \left(\frac{\partial x'^\mu}{\partial x^\nu}\right)$

Covariante:

$\mathcal{E}_{\mu\nu\rho\sigma}$ densidad tensorial de peso 1

$x^\mu \to x'^\mu$, la transformación tensorial estándar:

$$\mathcal{E}'_{\mu\nu\rho\sigma} = \underbrace{\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma}}_{\text{Por antisimetría es igual a } \det\left(\frac{\partial x}{\partial x'}\right) = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}} \mathcal{E}_{\alpha\beta\gamma\delta}$$

coordenadas nuevas arriba; $J = \left(\frac{\partial x'^\mu}{\partial x^\nu}\right)$

$$\mathcal{E}'_{\mu\nu\rho\sigma} = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)} \mathcal{E}_{\alpha\beta\gamma\delta}$$

y para mantener la normalización

$$\mathcal{E}'_{\mu\nu\rho\sigma} = \mathcal{E}_{\mu\nu\rho\sigma}$$

Significa que debemos agregar un factor $\det\left(\frac{\partial x'}{\partial x}\right)$

$$\mathcal{E}'_{\mu\nu\rho\sigma} = \det\left(\frac{\partial x'}{\partial x}\right)\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma}\mathcal{E}_{\alpha\beta\gamma\delta}$$

→ Para transformaciones que preservan la orientación $\left(\det\left(\frac{\partial x'}{\partial x}\right) > 0\right)$,

$|\det()| = \det()$

Comparando con (I):

$$\left|\frac{\partial x'}{\partial x}\right| = \det\left(\frac{\partial x'}{\partial x}\right) \quad \text{con } w = 1$$

⇒ $\mathcal{E}_{\mu\nu\rho\sigma}$ se transforma como una densidad tensorial covariante de peso 1.

• Una densidad tensorial $T_{\mu\nu\rho\sigma}$ de peso w se transforma bajo un cambio de coordenadas como:

$x^\mu \to x'^\mu$,

$$(I)\ldots T'^{\mu\nu\rho\sigma} = \left|\frac{\partial x'}{\partial x}\right|^w \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial x'^\rho}{\partial x^\gamma}\frac{\partial x'^\sigma}{\partial x^\delta} T^{\alpha\beta\gamma\delta}$$

$\left|\frac{\partial x'}{\partial x}\right|$ Valor absoluto

$\frac{\partial x'}{|\partial x|}$ es el valor absoluto del determinante de la matriz Jacobiana

$J = \left(\frac{\partial x'^\mu}{\partial x^\nu}\right)$

Contravariante:

$$\mathcal{E}'^{\mu\nu\rho\sigma} = \underbrace{\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial x'^\rho}{\partial x^\gamma}\frac{\partial x'^\sigma}{\partial x^\delta}}_{\text{Por antisimetría es igual a } \det\left(\frac{\partial x'}{\partial x}\right)} \mathcal{E}^{\alpha\beta\gamma\delta}$$

no hay necesidad de voltear nada. → $J = \left(\frac{\partial x'}{\partial x}\right)$

$$\mathcal{E}'^{\mu\nu\rho\sigma} = \det\left(\frac{\partial x'}{\partial x}\right)\mathcal{E}^{\alpha\beta\gamma\delta}$$

y para mantener la normalización

$$\mathcal{E}'^{\mu\nu\rho\sigma} = \mathcal{E}^{\mu\nu\rho\sigma}$$

$\frac{\partial x'}{\partial x}$ tiene más sentido físico (Jacobiano)

Significa que debemos agregar un factor $\det\left(\frac{\partial x}{\partial x'}\right) = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}$

$$\mathcal{E}'^{\mu\nu\rho\sigma} = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial x'^\rho}{\partial x^\gamma}\frac{\partial x'^\sigma}{\partial x^\delta}\mathcal{E}_{\alpha\beta\gamma\delta}$$

→ Para transformaciones que preservan la orientación $\left(\frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)} > 0\right)$,

$|\det()| = \det()$

Comparando con (I):

$$\left|\frac{\partial x'}{\partial x}\right| = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)} \quad \text{con } w = -1$$

⇒ $\mathcal{E}^{\mu\nu\rho\sigma}$ se transforma como una densidad tensorial covariante de peso −1.

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transformación de $\varepsilon^{\mu\nu\rho\sigma}$:

naturalmente como tensor de densidad

$$\varepsilon^{\mu'\nu'\rho'\sigma'} = \left|\frac{\partial x'}{\partial x}\right|^{w} \frac{\partial \mu'}{\partial \mu}\frac{\partial \nu'}{\partial \nu}\frac{\partial \rho'}{\partial \rho}\frac{\partial \sigma'}{\partial \sigma}\varepsilon^{\mu\nu\rho\sigma}$$

$$\Rightarrow \varepsilon'^{\mu\nu\rho\sigma} = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}\frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\varepsilon_{\alpha\beta\gamma\delta}$$

$$\Rightarrow g' = \det\left(\frac{\partial x'}{\partial x}\right)^{-2} g$$

$g<0$ $$\sqrt{-g'} = \det\left(\frac{\partial x'}{\partial x}\right)^{-1}\sqrt{-g}$$

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b) $\xi^{\alpha\beta\gamma\delta} = \sqrt{-g}\,\epsilon^{\alpha\beta\gamma\delta}$ es tensor?

$$\varepsilon'^{\mu\nu\rho\sigma} = \frac{\partial \mu'}{\partial \mu}\frac{\partial \nu'}{\partial \nu}\frac{\partial \rho'}{\partial \rho}\frac{\partial \sigma'}{\partial \sigma}\xi^{\alpha\beta\gamma\delta}$$

$$\xi^{\mu'\nu'\rho'\sigma'} = \sqrt{-g}\,\epsilon^{\alpha'\beta'\gamma'\delta'} = \left(\det\left(\frac{\partial x'}{\partial x}\right)^{-1}\sqrt{-g}\right)\frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}\frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\varepsilon^{\alpha\beta\gamma\delta}$$

$$= \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)^{2}}\frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\sqrt{-g}\,\varepsilon^{\alpha\beta\gamma\delta}$$

↖ No es un tensor.

$\xi^{\mu\nu\rho\sigma} = \sqrt{-\det(g_{\alpha\beta})}\,\epsilon^{\mu\nu\rho\sigma}$ = No es un tensor

↓ covariante ↘ contravariante — los componentes no se compensan

$\xi^{\mu\nu\rho\sigma} = \frac{1}{\sqrt{-\det(g_{\alpha\beta})}}\epsilon^{\mu\nu\rho\sigma}$ si sería tensor

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Cova:

$g_{\mu\nu}$: no es asimetrico $\neq \det\left(\frac{\partial x}{\partial x'}\right)$ (J)

$$g'_{\mu\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\mu}}\frac{\partial x^{\beta}}{\partial x'^{\nu}}g_{\alpha\beta}$$

$$= \left(J^{-1}\right)^{\alpha}_{\mu}\left(J^{-1}\right)^{\beta}_{\nu} g_{\alpha\beta}$$

⇝ forma matricial:

$$g'_{\mu\nu} = e'_{\alpha}\cdot e'_{\beta} = \left(\frac{\partial x^{\alpha}}{\partial x'^{\mu}}\hat{e}_{\alpha}\right)\cdot\left(\frac{\partial x^{\beta}}{\partial x'^{\nu}}\hat{e}_{\beta}\right)$$

$$= \frac{\partial x^{\alpha}}{\partial x'^{\mu}}\frac{\partial x^{\beta}}{\partial x'^{\nu}}(\hat{e}_{\alpha}\cdot\hat{e}_{\beta})$$

$$g'_{\mu\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\mu}}\frac{\partial x^{\beta}}{\partial x'^{\nu}}g_{\alpha\beta}$$

J_1^{-1} J_2^{-1} NO!

No hay múltiples Jacobianas diferentes para la misma transformación.

$\phi: M \to M$ la transformación es una función unica que mapea puntos del espacio y su J es derivada.

ϕ: $\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}r\\\theta\end{pmatrix}$ ↷ $\begin{pmatrix}r\cos\theta\\ r\sin\theta\end{pmatrix}$ $J = \begin{pmatrix}\frac{\partial\phi_1}{\partial r} & \frac{\partial\phi_1}{\partial\theta}\\ \frac{\partial\phi_2}{\partial r} & \frac{\partial\phi_2}{\partial\theta}\end{pmatrix}$

$J^{T} g_{\alpha\nu} J = \frac{\partial x}{\partial\mu}\frac{\partial x}{\partial\nu}g_{xx} + \dots$

⇝ $g_{\mu\nu}$ es simetrico

$$g'_{\mu\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\mu}}\frac{\partial x^{\beta}}{\partial x'^{\nu}}g_{\alpha\beta}$$

$$g'_{\alpha\beta} = \left(J^{-1}\right)^{T} g J^{-1}$$

~~$g'_{\alpha\beta} = \left(J^{-1}\right)^{T} g J^{-1} e_{\beta}$~~ def de inversion

matriz jacoviana inversa

$$\det(g'_{\alpha\beta}) = \det\left(\left(J^{-1}\right)^{T} g \left(J^{-1}\right)\right)$$

$$= \det\left(\left(J^{-1}\right)^{T}\right)\det(g)\det\left(J^{-1}\right)$$

$$= \det\left(J^{-1}\right)\det(g)\det\left(J^{-1}\right)$$

$$\det(g'_{\alpha\beta}) = \frac{1}{\det(J)}\det(g)\frac{1}{\det(J)}$$

$$\Rightarrow g' = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)^{2}} g$$

$$\Rightarrow g' = \det\left(\frac{\partial x'}{\partial x}\right)^{-2} g$$

Normalizemos $g'_{\mu\nu} = \det\left(\frac{\partial x'}{\partial x}\right)^{2}\frac{\partial x^{\alpha}}{\partial x'^{\mu}}\frac{\partial x^{\beta}}{\partial x'^{\nu}}g_{\alpha\beta}$ notion que $\left|\frac{\partial x'}{\partial x}\right|^{2}$ estar normalizado

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$$\Rightarrow \varepsilon'^{\mu\nu\rho\sigma} = \frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}\frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\varepsilon_{\alpha\beta\gamma\delta}$$

$$\Rightarrow g' = \det\left(\frac{\partial x'}{\partial x}\right)^{2} g$$

$g<0$ $$\sqrt{-g'} = \det\left(\frac{\partial x'}{\partial x}\right)^{1}\sqrt{-g}$$

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$$\xi^{\mu'\nu'\rho'\sigma'} = \sqrt{-g}\,\epsilon^{\alpha'\beta'\gamma'\delta'} = \left(\det\left(\frac{\partial x'}{\partial x}\right)\sqrt{-g}\right)\frac{1}{\det\left(\frac{\partial x'}{\partial x}\right)}\frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\epsilon^{\alpha\beta\gamma}$$

$$= \sqrt{-g}\,\frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\epsilon^{\alpha\beta\gamma\delta}$$

$$\xi^{\mu'\nu'\rho'\sigma'} = \frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\sqrt{-g}\,\epsilon_{\alpha\beta\gamma\delta}$$

$$\xi^{\mu'\nu'\rho'\sigma'} = \frac{\partial x^{\mu'}}{\partial x^{\alpha}}\frac{\partial x^{\nu'}}{\partial x^{\beta}}\frac{\partial x^{\rho'}}{\partial x^{\gamma}}\frac{\partial x^{\sigma'}}{\partial x^{\delta}}\xi^{\alpha\beta\gamma\delta}$$

↖ es un tensor.

$\xi^{\mu\nu\rho\sigma} = \sqrt{-\det(g^{\alpha\beta})}\,\epsilon^{\mu\nu\rho\sigma}$ = es un tensor los componentes se compensan

o tambien $\xi^{\mu\nu\rho\sigma} = \dfrac{1}{\sqrt{-\det(g_{\alpha\beta})}}\epsilon^{\mu\nu\rho\sigma}$

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o tambien $\xi^{\mu\nu\rho\sigma} = \dfrac{1}{\sqrt{-\det(g^{\alpha\beta})}}\epsilon^{\mu\nu\rho\sigma}$

Contra:

$g^{\mu\nu}$: no es asimetrico $\neq \det\left(\frac{\partial x}{\partial x'}\right)^{5}$

$$g'^{\mu\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g_{\alpha\beta}$$

$$= \left(J^{-1}\right)^{\mu}_{\alpha}\left(J^{-1}\right)^{\nu}_{\beta}\,g_{\alpha\beta}$$

⇝ forma matricial:

$$g'_{\mu\nu} = e'_{\alpha}\cdot e'_{\beta} = \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}}\hat{e}_{\alpha}\right)\cdot\left(\frac{\partial x'^{\nu}}{\partial x^{\beta}}\hat{e}_{\beta}\right)$$

$$= \frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}(\hat{e}_{\alpha}\cdot\hat{e}_{\beta})$$

$$g'_{\mu\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g_{\alpha\beta}$$

J_1 ↑ ↑ J_2 NO!

⇝ $g_{\mu\nu}$ es simetrico

$$g'_{\mu\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g_{\alpha\beta}$$

$$g'_{\alpha\beta} = \left(J\right)^{T} g\, J$$

~~$g'_{\alpha\beta} = \left(J^{-1}\right)^{T} g\, J^{-1} e_{\beta}$~~ (del de inversion)

matriz jacoviana inversa

$$\det(g'_{\alpha\beta}) = \det\left(\left(J\right)^{T} g\left(J\right)\right)$$

$$= \det\left(\left(J\right)^{T}\right)\det(g)\det(J')$$

$$= \det(J')\det(g)\det(J')$$

$$\det(g'_{\alpha\beta}) = \det(J)\det(g)\det(J)$$

$$\Rightarrow g' = \det\left(\frac{\partial x'}{\partial x}\right)^{2} g$$

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